

# Interval

## Literature evaluation & system proposal

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### Executive proposal

Interval is a working name for a from-scratch quiz portal intended for the sAIDE website at IIT Ropar. The project brief calls for a demonstration in at least two courses and a target of 350 simultaneous students, with later extension across departments. The proposed system supports graded and practice assessments on a responsive web interface. Its central design goal is to reduce avoidable coordination and interaction burden while keeping assessment rules, answer persistence and exceptions understandable.

This is an inception proposal, not a deployment or evaluation report. No local interviews, surveys, prototype tests, load tests or live-course pilots have been completed. The intended benefits and acceptance thresholds below are hypotheses and proposed targets. The name is provisional; brand or trademark clearance has not been undertaken.

### 1. Problem statement and evidence boundary

The team reports two current workflows: paper-and-pen quizzes that require classroom booking, and questions displayed on slides while students submit answers through Google Forms on mobile devices. The first has an explicit dependency on a room; the second splits question presentation and answer entry across surfaces. The team identifies cheating as a concern, but no incident rate or comparative effectiveness data has been supplied. Printing burden, reconciliation effort, marking delays and stress from interruptions are plausible investigation topics, not measured findings.

The design question is: how can one course-owned assessment workflow make quizzes easier to administer and complete without making ordinary device or connection failures look like misconduct? Moving online may reduce some room and paper dependencies, but it does not establish that unsupervised graded assessment is appropriate. The course owner must choose an assessment format and acceptable residual risk.

## 2. Method and scope

The review, conducted on 8 September 2026, uses official product documentation, two USENIX research papers, one explicitly labeled 2025 preprint, browser documentation, an accessibility reference and selected VIBE repository files. Four benchmarks represent different layers: Google Forms, Moodle, ClassMarker and Safe Exam Browser. This is a focused desk review, not a systematic literature review or hands-on usability comparison. No product accounts were created, and no paid-plan entitlements or prices were compared.

Course grounding comes from the supplied AI511 lectures on interaction design, its process, conceptual models, cognitive aspects and interfaces, together with the supplied fifth-edition Interaction Design textbook [C1]. In particular, the proposal separates assumptions from claims, uses provisional personas, and maps design decisions to feedback, consistency, error recovery and recognition over recall.

## 3. Provisional users and design goals

Student: complete a quiz under understandable rules and distinguish pending work from saved answers and final submission. Instructor: create, preview, publish, grade and release results for the correct course. TA/administrator: resolve interrupted attempts, manage permissions and inspect a scoped incident history. These are role-based proto-personas inferred from the task, not personas validated through participant research.

Proposed usability goals are task effectiveness, learnability, clear system status and recoverability. Proposed experience goals are confidence in submission, comprehensible rules and reduced unnecessary monitoring burden. Accessibility is part of the core interaction design rather than a late feature: keyboard operation, readable responsive layouts, non-colour-only status and suitable timing accommodations are explicit design requirements. WCAG 2.2 is a design reference, not a certification claim [S11].

# Part A · Literature and implementation gaps

## 4. Benchmark evaluation

Google Forms supports answer keys, grading, feedback and controlled grade release [S1]. Its documented locked mode requires school-managed Chromebooks and a Google Workspace for Education account [S2]. This is a concrete constraint when the project asks for mixed personal-device access; it is not evidence that ordinary Forms cannot be used on mobile. A local workflow that projects questions separately also need not be a limitation of Forms itself: moving question content into the form is a credible baseline to compare against.

Moodle 5.1 documents timed quizzes, automatic handling at expiry, user/group overrides, navigation choices and result-review controls [S3]. It is a serious alternative, not an intentionally weak comparator. No local evidence currently demonstrates that Moodle is too difficult, too slow or unsuitable. A guided course-specific workflow is a design hypothesis to test against it. Version 5.1 identifies the documentation reviewed; it is not a claim about the latest release.

ClassMarker documents question banks, randomization, timing and results export [S4]. Its browser-monitoring workflow already warns on focus exits and makes recorded events available for review [S5]. Consequently, tab monitoring and event review cannot be presented as novel features. Local policy ownership, integration, support arrangements and suitability for the specific courses still require investigation; lack of documentation in this review is not treated as feature absence.

Safe Exam Browser is a locally installed assessment environment that connects to a quiz/LMS service and restricts access to other resources [S6]. It occupies a different layer from authoring and grading. Its documented platforms are Windows, macOS and iOS; the project must verify actual student-device coverage before proposing it as an optional stricter mode. An ordinary browser portal should not claim the same environment control as a managed client.

## 5. Literature evaluation

Balash et al. (SOUPS 2021) analyse browser-extension reviews and survey 102 student test-takers [S7]. Privacy concerns and institutional power dynamics make a meaningful case for transparent preflight information and minimum necessary monitoring. The sample is US-based and self-reported, so the study does not establish the prevalence of these concerns at IIT Ropar. It is not an experiment measuring this portal's anti-cheating effectiveness.

Balash et al. (USENIX Security 2023) survey 125 educators at one US institution [S8]. Their findings describe varied proctoring decisions, alternative assessment choices and privacy trade-offs. This supports treating integrity policy as an explicit instructor decision rather than imposing one monitoring setting on practice and graded work. Single-institution recruitment, limited participation and self-report restrict generalization; the paper does not prove that a specific configuration prevents cheating.

Kwapisz et al. (2025 preprint) report 16 semi-structured interviews with students using disability accommodations [S9]. The analysis connects surveillance anxiety, fear of misrepresentation and cognitive burden. It motivates testing accommodations, predictable warnings and a human-review route. This qualitative work is not a prevalence estimate, and the authors caution against generalizing across diverse lived experiences. The source is labeled as a preprint, not assumed to be peer-reviewed.

## 6. Defensible gap statement

The strongest gaps are contextual: fragmented local workflows, unclear persistence and submission states, poorly explained enforcement consequences, and the need for one accountable cross-course service. Some are locally reported; others remain hypotheses. The literature adds evidence of privacy and accessibility tensions in monitored assessment. The review does not demonstrate a universal absence of features across existing systems. Interval's contribution is a locally testable design and implementation under the course's from-scratch requirement, not a claim to have invented quiz infrastructure.

## Part B · Proposed solution and architecture

### 7. Gap-to-design traceability

G1: split question and answer surfaces. Proposed response: display the question, answer controls and progress within one attempt. HCI rationale: consistency and reduced cross-surface coordination. Test: observe task completion and answer-entry mistakes against the actual current workflow.

G2: uncertainty about whether an answer is saved or a quiz is finished. Proposed response: distinct pending, saving, saved and submitted states, plus a server-confirmed receipt. HCI rationale: visibility and feedback. Test: ask participants to identify the state and verify acknowledged answers through refresh and connection interruption.

G3: opaque or disproportionate integrity enforcement. Proposed response: preflight policy disclosure, configurable off/warn/strict behaviour, recorded reasons and authorized review. HCI rationale: understandable constraints and error recovery. Test: policy comprehension, ordinary interruption scenarios and exception resolution. The distinction between page visibility and focus in browser documentation matters here [S10].

G4: risky setup or cross-course administration. Proposed response: draft, preview and publish stages; course-scoped access; versioned rules and grading changes. HCI rationale: recognition over recall, error prevention and consistent conceptual models. Test: instructors publish to the intended cohort and explain the student-visible policy without assistance.

## 8. Architecture and technical choices

Proposed flow: responsive browser interface → authenticated application service → relational database. A background worker handles expiry finalization, grading work and exports outside the interactive save path. The architecture is a modular application, not a collection of microservices. The school website links or routes to the app; an identity adapter separates institutional authentication from quiz logic.

A reasonable implementation candidate is a TypeScript/React interface, a TypeScript application API and PostgreSQL, subject to hosting and team review. These are proposals, not discoveries about the existing school website. The public site confirms the integration context, but it does not reveal approved runtime support, administrator access, an SSO contract or a functioning /quiz route [S16]. Do not publish a deployment promise before these are confirmed.

Core entities are Course, Membership, QuizVersion, QuestionVersion, Attempt, AnswerRevision, PolicyEvent, ReviewDecision and Result. A published quiz freezes its applicable questions and rules. Each attempt receives a server-assigned question order and deadline. Authorization checks protect each course and attempt, rather than trusting a course identifier supplied by the browser.

The server decides whether an answer may be accepted, whether an attempt is locked or expired, and which marks and answer keys may be released. The client displays these authoritative states. A save includes a revision identifier; acknowledgement follows durable storage. Repeated save or submit requests must not produce duplicate final records. Submission finalization and its receipt should be transactional. Scheduled reconciliation must finish expired attempts even if the student's browser disappears. Automatic grading must not expose answer keys in an unreleased attempt response.

A network outage is distinct from a focus event. Unsynchronised edits are explicitly pending, not saved. The proposed initial deadline policy accepts answers that arrive at the server by the deadline; pending offline edits that arrive later are not silently accepted based on a client clock. The system preserves acknowledged work, shows the limitation before the attempt, and routes exceptional cases to an authorized instructor. A more generous grace policy would need an explicit threat model and separate design decision.

## 9. Configurable integrity policy

Off: no focus-event collection. Warn and record: log supported events and durations, explain the event and allow the attempt to continue. Strict: on the configured supported trigger, request a server-side attempt lock and quiz-session logout; preserve acknowledged answers, keep the deadline running and permit re-entry only after an authorized decision. Strict affects the attempt, not unrelated institutional sessions. The instructor selects the trigger and consequence in advance; settings should not silently change mid-attempt.

Page visibility and focus are different signals [S10]. Neither proves where a student looked, what another device displayed or whether misconduct occurred. Event delivery can fail, and client-side reporting is not a tamper-proof security boundary. Therefore immediate tab-exit logout is a best-effort policy on ordinary browsers, not guaranteed detection. The UI should say what happened, what was saved and how to request review; it should not announce that the student has been “caught cheating.”

Randomization, well-designed question banks, appropriate resource rules and human review complement policy controls. Reduced supervision is an aspiration to validate, not a technical guarantee. Camera, face, gesture or voice monitoring is an optional later track. It requires a separate accuracy and false-flag evaluation, device testing, an approved data-handling policy and a meaningful alternative where monitoring is unsuitable. Emotion inference and automatic misconduct verdicts are outside the proposed core.

## 10. Feature roadmap and prior experience

Lab 5: design an instructor publish task and a student complete-and-submit task. Produce low-fidelity flows, key interaction screens and a clickable prototype in the next phase. This Lab 4 package does not claim to contain that prototype.

Lab 6 (23 September 2026): implement a baseline vertical slice—one MCQ quiz with course access, a server deadline, acknowledged saves and a final receipt. Labs 7–9: test, improve and validate that slice. Labs 10–12: complete the agreed feature set and release only after acceptance gates are met. Dates beyond those in the supplied outline are not assigned.

Full proposed scope: single/multiple choice, numeric and short-answer questions; banks and imports; media and mathematical notation; timing and shuffling; attempt limits and review flags; answer recovery; auto/manual grading and optional negative marks; controlled result release, CSV export and item analysis; course roles, audit trails and accommodations; configurable integrity policy and review. Optional AI monitoring is gated separately. “All features” is treated as a roadmap, not a claim that all are feasible in the initial milestone.

Saaransh reports prior work on VIBE, including AI-proctored quizzes. Public code inspection finds attempt/save/submit hooks and result-display controls in quiz.tsx [S13], detector configuration in EditProctoringModal.tsx [S14], and policy/reinstatement hooks [S15]. The README describes the wider learning and monitoring concept [S12]. This supports relevance of the prior project, not personal authorship of individual changes or detector effectiveness. The inspected proctoring modal disables its blur checkbox in that UI; no repository-wide behavioural conclusion is drawn. Interval remains a new implementation; VIBE code has not been copied into it.

## 11. Proposed evaluation and release gates

**Primary research:** seek voluntary input from students in the two intended courses, instructors/TAs and the website administrator. Use neutral last-experience questions, not prompts that assume paper is bad or proctoring is desirable. Recruit across device types and invite accessibility needs without collecting diagnoses. The accompanying research kit provides questions and an empty evidence structure; it contains no fabricated responses.

**Usability:** observe whether a student can complete and submit an attempt and whether an instructor can publish a correctly configured quiz. Record completion, assistance, mistakes, policy comprehension, confidence in save status and time on task. Suggested formative recruitment is 6–8 students and 2–3 instructors/TAs; this is a proposed practical sample, not a statistically representative one. Report raw counts and contexts rather than inflated percentages.

**Reliability:** model 350 concurrent sessions with realistic question content and traffic. An illustrative save cadence of one batched request per student every five seconds gives  $350/5 = 70$  save requests per second before other traffic. Test simultaneous starts and submissions, retries, refreshes, disconnections and duplicate requests. Proposed gates are p95 save acknowledgement below one second in the agreed workload, no loss of acknowledged answers, and one durable final receipt per completed attempt. Hardware, duration, payload sizes and network assumptions must accompany results. No capacity result currently exists.

**Accessibility and safety:** test keyboard-only use, readable mobile layouts, screen-reader feedback, non-colour status cues and timing overrides. Verify unauthorized cross-course access is denied, answer keys remain unreleased, and policy locks have a documented review route. Do not count a generated incident flag as proven misconduct. Do not claim WCAG conformance without a full evaluation.

**Live pilot:** secure approval for two courses, confirm support and device access, and rehearse a no-penalty fallback before a graded use. A practice rehearsal precedes the graded pilot. Preserve the last acknowledged state during failure. Measure administrative effort, room bookings and paper use against an observed baseline if reduction claims are made. Do not infer paper, time or cost savings from the existence of a website.

## 12. Risks, ownership and unresolved decisions

The major risks are overload at start/submit peaks, unreliable connections, false-positive enforcement, inaccessible device requirements, unauthorized data access and insufficient maintenance ownership. Mitigations are load/fault testing, honest pending-state feedback, minimum necessary monitoring, recovery pathways, role checks and a documented support plan. From-scratch development creates ongoing security, deployment and support work; it is not automatically the lowest-cost institutional choice.

Still to confirm: two pilot-course identities and approvals; website hosting and access; institutional login mechanism; supported browser/device matrix; question and grading semantics; retention and review ownership; and policies for network failure, strict-mode exceptions and accommodations. Proposed technology and thresholds must not be confused with approved requirements. There is no legal-compliance or production-readiness certification in this report.

## Conclusion

The case for Interval is not that quiz features are missing from the market. It is that this project can design, build and evaluate one coherent local assessment journey with explicit rules, recoverable work and accountable administration. The next milestone should demonstrate two task paths, not a wall of features. Trust should be earned through observed usability and reliability before the portal carries graded work at scale.

## References

All web sources were checked on 8 September 2026. Full URLs, review scope and source limitations are recorded in `Research_Notes.md` alongside this report.

[S1] Google. Create & grade quizzes with Google Forms. Google Docs Editors Help. [support.google.com/docs/answer/7032287](https://support.google.com/docs/answer/7032287)

[S2] Google. Use locked mode for quizzes. Chrome Enterprise and Education Help. [support.google.com/chrome/a/answer/7634943](https://support.google.com/chrome/a/answer/7634943)

[S3] Moodle. Quiz settings. MoodleDocs, version 5.1. [docs.moodle.org/501/en/Quiz\\_settings](https://docs.moodle.org/501/en/Quiz_settings)

[S4] ClassMarker. User manual. [classmarker.com/online-testing/manual/](https://classmarker.com/online-testing/manual/)

[S5] ClassMarker. Automated Exam Proctoring using Browser monitoring. [classmarker.com/online-testing/manual/classmarkermonitor](https://classmarker.com/online-testing/manual/classmarkermonitor)

[S6] Safe Exam Browser. About Overview. [safeexambrowser.org/about\\_overview\\_en.html](https://safeexambrowser.org/about_overview_en.html)

[S7] Balash, D. G., Kim, D., Shaibekova, D., Fainchtein, R. A., Sherr, M., and Aviv, A. J. (2021). Examining the Examiners: Students' Privacy and Security Perceptions of Online Proctoring Services. Seventeenth Symposium on Usable Privacy and Security (SOUPS 2021), 633–652. USENIX Association.

[S8] Balash, D. G., Korkes, E., Grant, M., Aviv, A. J., Fainchtein, R. A., and Sherr, M. (2023). Educators' Perspectives of Using (or Not Using) Online Exam Proctoring. 32nd USENIX Security Symposium, 5091–5108. USENIX Association.

[S9] Kwapisz, M. B., Ackerman, Y., Nguyen, J., and Rajivan, P. (2025). Surveillance and Disability in Online Proctored Exams: Student Perspectives and Design Implications. arXiv:2511.10826v1. Preprint. [arxiv.org/abs/2511.10826](https://arxiv.org/abs/2511.10826)

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[S11] W3C Web Accessibility Initiative. How to Meet WCAG: WCAG 2.2 Quick Reference. [w3.org/WAI/WCAG22/quickref/](https://w3.org/WAI/WCAG22/quickref/)

[S12–S15] vicharanashala/vibe. Public GitHub repository, main branch: README.md; frontend/src/components/quiz.tsx; frontend/src/components/EditProctoringModal.tsx; frontend/src/hooks/ejection-policy-hooks.ts. Selected source review, not a runtime audit. [github.com/vicharanashala/vibe](https://github.com/vicharanashala/vibe)

[S16] sAIDE, IIT Ropar. Public school website. [saide.iitrpr.ac.in](https://saide.iitrpr.ac.in)